



muscle relaxants, and then reused to vaccinate the children. In such a case it can kill instantly. Alternatively, the deaths could have taken place because the vaccine, after being reconstituted, was kept out in the open, causing bacterial growth and contamination.

Protocols are routinely followed like checking for the indicators and discarding vials when there is a colour change. But the possibility of a callous attitude in taking what was the lethal dose from used vials cannot be ruled out as they are kept in the same ice-lined refrigerator for a week for testing. However, holding health workers alone responsible, without assessing the underlying causes like lack of good infrastructure and emergency response systems to handle adverse situations, may harm rather than strengthen the immunisation programme.

The irony is that HBI is a first time supplier of the measles vaccine to the Health Ministry. At Rs 39 for a vial of five doses HBI's bid was Rs 4 less than what was quoted by its competitor. Earlier, the company has supplied anti-rabies, Hepatitis-B, DPT, DT and TT vaccines. It was supplying ten lakh doses of anti-rabies vaccine every year to Tamil Nadu alone. "Till today our vaccines have never had any problems," says HBI General Manager Dinar A. Kumar adding that the company is a WHO-certified vaccine maker, which follows all manufacturing procedures, storage and transport practices as well as quality norms scrupulously.

FOR GOOD HEALTH: An infant being vaccinated for measles

FOR THE DEATHS AS FOUR CHILDREN DIED IN QUICK SUCCESSION." DR. SURINDER SINGH, DRUGS CONTROLLER OF INDIA

centres in the state have generators for backup in case of power cuts and the temperature under which the vaccines are kept is also monitored continuously. Poor cold storage facilities and lack of hygienic needles could make the vaccine toxic and deadly. Besides the right temperatures for storage, it is also essential that the vaccine is used within three hours of reconstitution. "A bad vaccine is highly unlikely because they are tested at CRI before being released and the expiry date of this batch was 2010. They also had vial monitors, which change colour if a vaccine goes

bad. A human error looks to be the most likely cause here. Contamination of the needle used could be a possibility. Four children dying within an hour of being vaccinated is a direct result of an anaphylactic shock—a severe, acute, potentially lethal systemic allergic reaction. Such a reaction following a measles vaccination is rare and occurs in one out of every 10 lakh cases," explains the Drugs Controller of India Surinder Singh.

Vaccine experts are apprehensive of the syringe having been dipped in other chemicals, like adrenaline or

While the findings of the inquiry may finally nail down the causes and the people responsible for the unfortunate deaths, the manner in which they have occurred is likely to affect the Government's immunisation programme as apprehensive parents may steer clear from any concerted drive in the near future. As it is, India falls in the list of the top six countries with high infant deaths. Ways will have to be found to restore people's faith in the programme and maintain wide vaccination coverage to protect children and ensure safe motherhood. ■